

ANALYSIS OF THE CASH FLOW STATEMENT: REFORMULATIONS

Chapter 10



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- ☐ REFORMULATION OF THE CASH FLOW STATEMENT: SOME ALTERNATIVES
- ☐ FROM CFOA TO OFCF
- ☐ STANDARD VS. REFORMULATED CFS (CHANGE IN “CASH”)
- ☐ WHAT TO DO WITH....UNUSUAL ITEMS
- ☐ LET’S SEE WHAT TO DO WITH PROVISIONS EXPENSES
- ☐ AND NOW WE CAN UNDERSTAND!
- ☐ CHANGING THE BOTTOM LINE : “CASH” VS. “NET FINANCIAL POSITION”
- ☐ FOCUS ON THE CHANGE IN «NET FINANCIAL POSITION»
- ☐ ATTENTION, PLEASE...
- ☐ THE TWO TYPICAL «FINANCE» VIEWS OF CASH FLOWS

Cash Flow Statements prepared according to Accounting Standards (“standard” cash flow statements) explain the change in “cash & cash equivalents” and they classify cash flows into 3 categories: Operating, Investing and Financing Activities

Reformulated cash flow statements can present many differences:

1. they can explain, as final result (bottom line), **a different type of “financial resources”**:
 - a) change in “Cash & Cash equivalents”
 - b) change in “Net Financial Position”
 - c) change in “Cash Flow to Equity” (available to shareholders)
2. they usually classify cash flows in order to calculate the **Operating Free Cash Flow** (separately from “**Other Cash Flows**”)
3. The most common reformulations use to **reconcile the Operating Income with the related cash flow**:
 1. Indirect method (starting from the **Operating Income**)
 2. A concept of working capital according to the “current” criteria, **NOWC** (non the “short-term” criteria)
 3. Measurement of both the **Operating Cash Flow** and the **Operating Free Cash Flow** (OCF – CAPEX)



The Standard Cash Flow Statement highlights the “**Cash Flow from Operating Activities**” (CFOA), i.e. the cash flow generated by all the income statement items (that’s why, using the indirect method, the starting point is the “net income”)

The **Operating Free Cash Flow (OFCF)** is the difference between

- **Operating Cash Flow**, i.e. the cash flow generated only by the “operating” income statement items (that’s why, using the indirect method, the starting point is the “operating income”)
- **CAPEX** (net investments in Non Current Operating Assets. i.e. tangible and intangible assets)

In the next slide you will see how to move

- from the «Standards» Cash Flow Statement (i.e prepared according to the Accounting Standards)
- to a «Reformulated» Cash Flow Statement where the Free Operating Cash Flow is separated from the “other (non operating) cash flows”,

given the same bottom line, i.e. the Net Change in Cash and Cash Equivalents.

With the help of some colors you will see how to move the various elements.

“Standard” Statement of Cash Flows

➤ +/- Operating Cash Flow
➤ + Interest and dividend received
➤ - Interest paid
➤ - Income taxes paid
+ / - “Cash Flow from Operating Activities”
➤ -/+ Purchase(Sale) of tangible and intangible assets
➤ -/+ Purchase(Sale) of Financial Investments
- / + “Cash Flow from Investing Activities”
➤ + Borrowings from financial institutions
➤ - Reimbursements to financial institutions
➤ +/- Issuing(Reimbursement) of debt securities (bonds)
➤ + Issuing equity securities
➤ - Dividend paid
+ / - “Cash Flow from Financing Activities”

= Net Change in Cash and Cash Equivalents**And...what to do if you want to separate Unusual Items?**

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Reformulated Statement of Cash Flows

➤ +/- Operating Cash Flow
➤ -/+ Purchase(Sale) of tangible and intangible assets (CAPEX)
+ / - “Operating Free Cash Flow” (OFCF)
➤ + Interest and dividend received
➤ - Interest paid
➤ - Income taxes paid
➤ -/+ Purchase(Sale) of Financial Investments
➤ + Borrowing cash from financial institutions
➤ - Reimbursements to financial institutions
➤ +/- Issuing(Reimbursement) of debt securities (bonds)
➤ + Issuing equity securities
➤ - Dividend paid
+/- “Other Cash Flows” (non operating)

= Net Change in Cash and Cash Equivalents

This is a complex problems.....because if we separate Unusual Items from the Operating Income, we should theoretically do the same in both:

- ☐ the Balance Sheet (unusual assets, liabilities and OCI)
- ☐ the Cash Flow Statement (unusual cash flows)

This way we could put a “Sustainable (Recurring) Operating Cash Flow” beside the “Sustainable (Recurring) Operating Income”, because of the coherence among the three financial statements:

1) Income Statement – distinction between:

- Operating Income = Recurring Operating Income
- Unusual Items = Infrequent, Anomalous (in nature), related to previous year etc.....

2) Cash Flow Statement – distinction between:

- Operating Cash Flow = Recurring Operating Cash Flow
- Cash Flow from Unusual Items (classified among the “Other Cash Flows” items)

3) Balance Sheet – distinction between:

- Current Operating Assets and Liabilities (NOWC) = Assets and Liabilities related to re-current Operating revenues and Expenses (our Recurring Operating Income)
- Assets and liabilities related to Unusual Items (if some unusual items were originally classified into the reported Operating Income we must exclude the related assets and liabilities from the NOWC...)



Is it possible to identify the B/S and CF/S Unusual Items (so that they can be combined with the I/S Unusual Items)?

Sometimes yes, sometimes it's difficult (because of a poor disclosure in the Notes to Financial Statements).....

Some “easy” examples:

- a) “unusual” provision expenses for litigation (I/S); related change in “unusual” provision liability (B/S) → “unusual” payment of litigation settlements (CF/S);
- b) “unusual” restructuring expenses (IS); related change in “unusual” provision liability” (B/S) → “unusual” payment of restructuring expenses (CF/S);

Some “non easy” examples:

- a) “unusual” merge expenses (I/S); related change in “unusual” accounts payable (?) (B/S) → “unusual” payment of merge expenses (?) (CF/S);
- b) “unusual” FV adjustment (O.C.I.) of Cash Flow Hedging IRS (I/S); related changes in A.O.C.I. of accumulated reserves for IRS's FV adjustments; → “unusual” periodic settlement cash received or paid (?) (CF/S);
- c) “unusual” translation adjustment (O.C.I.) for foreign subsidiaries (I/S); “unusual” change in translation adjustment reserve in A.O.C.I. (B/S) → (?) (is there any cash flow?)

Note: (?) = possible missing item

Remember: for some “unusual items” there are no cash flows: ex. Fair Value adjustments, revaluations, impairments, changes in estimates because of changes in Accounting Standards...

Remember: for “unusual items” like gains/losses from disposals of assets the related cash flow is usually separately disclosed in the Cash Flow Statement (“Investing Activities” section)

Let's suppose that a company reports two provision expenses and liabilities:

- 1) for warranties (to customers);
- 2) for litigation because of an intellectual property breach.

The analyst adopts the following classification choices:

1) Warranties

- ☐ It's a recurring cost (being the result of frequent sales to customers) → it affects the EBITDA
- ☐ The provision liability is classified as current operating liability → it affects the NOWC
- ☐ The Operating Cash Flow (EBITDA $-/+ \Delta$ NOWC) includes any payment for warranties

2) Litigation

- ☐ It's an unusual cost (being the result of infrequent events) → **NO**: EBITDA
- ☐ The provision liability is classified as "other liability" in order not to contaminate the current operating liabilities and, consequently, the NOWC → **NO**: NOWC

Let's see an example with numbers



Reported Income Statement

Revenues	60.000
.....	
Warranties expenses (260), net of provision utilization (220)	-40
Litigation expenses (450), net of provision utilization (150)	-300
Provision expense for warranties	-250
Provision expense for litigation	-5.000
.....	
EBITDA	4.500
Depreciation and Amortization	-800
Operating Income	3.700

Reformulated Income Statement

Revenues	60.000
.....	
Warranties expenses (260), net of provision utilization (220)	-40
Provision expense for warranties	-250
.....	
EBITDA	9.800
Depreciation and Amortization	-800
Operating Income	9.000
Litigation expenses (450), net of provision utilization (150)	-300
Provision expense for litigation	-5.000
Unusual items	-5.300

Reformulated Balance Sheet

	beg	end	change
.....			
Provision liability for warranties	380	410	30
NOWC	4.600	5.200	600
Non Current Operating Assets	45.000	52.000	7.000
TOTAL OPERATING ASSETS	49.600	57.200	7.600
Provision liability for litigations	150	5.000	4.850
Other Liabilities	150	5.000	4.850
Financial Liabilities	15.200	16.000	800
Net Equity	34.250	36.200	1.950
TOTAL LIABILITIES AND EQUITY	49.600	57.200	7.600

Warranties expenses (260), net of provision utilization (220)	-40 EBITDA
Provision expense for warranties	-250 EBITDA
Increase in Provision Liability	30 ΔNOWC
Paid Warranties Expenses	-260

The Operating Cash Flow includes the amount of paid warranty expenses

Reformulated Cash Flow Statement

EBITDA (reformulated)	9.800
- Increase of NOWC	-600
Operating Cash Flow	9.200

Which is the amount of paid warranty expenses?

Litigation expenses (450), net of provision utilization (150)	-300 Unusual Items (I/S)
Provision expense for litigation	-5.000 Unusual Items (I/S)
Increase in Provision Liability	4.850 ΔOther Liabilities
Paid Litigation Expenses	-450

Other Cash Flows:

.....	
Payment of Litigation Settlements	-450
Cash Flow for Unusual items	-450

How can you get this cash flow?



- ❑ In Session 03 (slides 8-9) we said:
- ❑ «Provisions do not represent potential cash outflows in the short-term future (“cash flow view” of EBITDA); but this is true only for long-term provisions.....”

- ❑ Now we can better clarify the concept!
- ❑ In coherence with the use of both
 - the “managerial” B/S (current vs. other liabilities) and
 - the reformulated I/S with the distinction between the Recurring Operating Income and the Unusual Items
- ❑ The EBITDA (but also the Operating Income!) must exclude those provision expenses that are classified as unusual items and, therefore, do not affect the current liabilities (but the “other liabilities”)
- ❑ **Trick:** if you classify “unusual” provision liabilities as Current Operating Liabilities (and not as “Other Liabilities”, because you don’t adopt this category), remember to eliminate them from the NOWC when you calculate the “recurring” Operating Cash Flow

= EBITDA

- Depreciation and Amortization, - Impairments (+Reversals) and - **(LT) provision expenses**

= Core Operating Income (from sales of products/services)

= EBITDA **(including recurring provision expenses)**

- Depreciation and Amortization, Impairments/Reversals

= Core Operating Income (from sales of products/services)

.....

.....

= Earnings Before Unusual Items and Taxes

+/- Unusual Items **(including unusual provision expenses)**

Earnings Before Taxes

.....



Reformulated CFS of the change in Cash

+/- Operating Cash Flow

-/+ Purchase(Sale) of tangible and intangible assets (CAPEX)

+/- “Operating Free Cash Flow” (OFCF)**1) Cash Flows from financial income/expenses , unusual items and income taxes:**

+ Interest and dividend received

- Interest paid

+/- Unusual Items received/paid

- Income taxes paid

2) Cash Flows from Financial Investing Activities:

-/+ Purchase(Sale) of Short-term Financial Investments

-/+ Purchase(Sale) of Long-term Financial Investments

3) Cash Flows from Financing Activities (with third parties):

+/- Borrowings(reimbursements) oh Short-term loans

+/- Borrowings(reimbursements) oh Long-term loans

4) Cash Flows from Financing Activities (with shareholders):

+ Issuing equity securities

- Dividend paid

+/- “Other Cash Flows”**= NET CHANGE IN CASH AND CASH EQUIVALENTS**

Reformulated CFS of the change in NFP

➤ +/- Operating Cash Flow

➤ -/+ Purchase(Sale) of tangible and intangible assets (CAPEX)

+/- “Operating Free Cash Flow” (OFCF)**1) Cash Flows from financial income/expenses , unusual items and income taxes:**

+ Interest and dividend received

- Interest paid

+/- Unusual Items received/paid

- Income taxes paid

2) Cash Flows from Financial Investing Activities:**NO (their change is included into the Net Financial Position)**

-/+ Purchase(Sale) of Long-term Financial Investments

3) Cash Flows from Financing Activities (with third parties):**NO (their change is included into the Net Financial Position)****NO (their change is included into the Net Financial Position)****4) Cash Flows from Financing Activities (with shareholders):**

+ Issuing equity securities

- Dividend paid

+/- “Other Cash Flows”**= NET CHANGE IN NET FINANCIAL POSITION**

Reformulated CFS of the change in Cash

+/- Operating Cash Flow

-/+ Purchase(Sale) of tangible and intangible assets (CAPEX)

+/- “Operating Free Cash Flow” (OFCF)

1) Cash Flows from financial income/expenses, unusual items and

In the Reformulated Statement of change in Net Financial Position the “Other Cash Flows” exclude the cash flows related to those balance sheet accounts that are included into the Net Financial Position:

- short-term and long-term financial liabilities
- short-term financial assets

My rule: “if something is inside the bottom line it can’t stay above the bottom line”

4) Cash Flows from Financing Activities (with shareholders):

+ Issuing equity securities

- Dividend paid

+/- “Other Cash Flows”

= NET CHANGE IN CASH AND CASH EQUIVALENTS

Reformulated CFS of the change in NFP

➤ +/- Operating Cash Flow

➤ -/+ Purchase(Sale) of tangible and intangible assets (CAPEX)

+/- “Operating Free Cash Flow” (OFCF)

1) Cash Flows from financial income/expenses, unusual items and income taxes:

+ Interest and dividend received

- Interest paid

+/- Unusual Items received/paid

- Income taxes paid

2) Cash Flows from Financial Investing Activities:

NO (their change is included into the Net Financial Position)

-/+ Purchase(Sale) of Long-term Financial Investments

3) Cash Flows from Financing Activities (with third parties):

NO (their change is included into the Net Financial Position)

NO (their change is included into the Net Financial Position)

4) Cash Flows from Financing Activities (with shareholders):

+ Issuing equity securities

- Dividend paid

+/- “Other Cash Flows”

= NET CHANGE IN NET FINANCIAL POSITION

Attention to what (and how) impacts on the Net Financial Position (3 cases):

1. **change of the NFP's amount because of "cash transactions" (from the Reformulated Cash Flow Statement):**
 - a) Operating Free Cash Flow
 - b) Other Cash Flows (from financial income/expenses, unusual items and income taxes; from long-term financial investing activities; from financing activities with shareholders)
2. **change of the NFP's content (but not of the NFP's amount) because of "internal cash transactions":**
 - a) Borrowings and reimbursements (in cash) of short-term and long-term financial liabilities
 - b) Purchase and sale (in cash) of short-term securities
 - c) Loans and reimbursement (in cash) of short-term financial receivables
 - d) change in "cash and cash equivalents" (related to the above mentioned transactions)
3. **change of the NFP's amount because of "accounting valuations" and "non cash transactions":**
 - a) Change in valuation of financial liabilities (because of: amortized cost, fair value)
 - b) Change in Fair Value and impairments(reversals) of short term financial assets (securities and financial receivables)
 - c) Non cash transactions that affect the NFP components: ex. increase of long-term financial liabilities for lease of PPE, exercise of the conversion option for convertible bonds etc.

The reformulated cash flow statement always presents a **reconciliation** of the overall change in Net Financial Position

1. *Cash Flow from Operating Activities* $\neq$ *Operating Cash Flow*
2. And what is the *Cash Flow from Operations*? can be either the Cash Flow from Operating Activities or the Operating Cash Flow....
3. **OFCF** can be calculated *before or after taxes* (starting from NOPAT)
4. **OFCF after taxes** is also called “*Unlevered Free Cash Flow*” or “*Free Cash Flow to Firm*” or, simply, “*Free Cash Flow*”
5. **Sometimes**: OCF include interest and dividend received....attention! (it happens when the Operating Income includes Financial Income)
6. **Sometimes**: Free Cash Flow = Cash Flow from Operating Activities – CAPEX...
7. And what about the **Free Cash Flow to Equity**? ➔ See next slide

Unlevered Free Cash Flow, i.e OFCF after taxes (usually discounted by using wacc)

-/+ Net Financial Expenses, net of taxes

+/- Net change in Net Financial Obligations

= **Free Cash Flow to Equity** (discounted by using the cost of equity, K_e)

+ Issuance of equity securities

- Payment of dividends

- Repurchase of equity shares (including the purchase of treasury stock)

= **Net change in Cash & Cash Equivalents**



Analysis of **change in Net Financial Position**,
with detail of **Operating Free Cash Flow**

Analysis of **Free Cash Flow to Equity**, with
detail of **Operating Free Cash Flow**

the same cash flows

+EBITDA

+/-change in Net Operating Working Capital

= Operating Cash Flow

- Net investments in long-term operating assets

= Operating Free Cash Flow (before taxes)

+ Interest and dividends received

- Interest paid

- Income taxes paid

- Net investments in long-term financial assets

- Dividends paid

+/- Change in Stockholders' equity

= Change in Net Financial Position, of which:

- ☐ - Reimbursement of financial debts
- ☐ + Proceeds from borrowings
- ☐ -/+ Change in short-term financial assets
- ☐ +/- Change in Cash & cash Equivalents

+EBITDA

+/-change in Net Operating Working Capital

= Operating Cash Flow

- Net investments in long-term operating assets

= Operating Free Cash Flow (before taxes)

+ Interest and dividends received

- Interest paid

- Income taxes paid

- Net investments in long-term financial assets

- Reimbursement of financial debts

+ Proceeds from borrowings

-/+ Change in short-term financial assets

= FCFE (Free Cash Flow to Equity), of which:

- ☐ - Dividends paid
- ☐ +/- Change in Stockholders' equity
- ☐ +/- Change in Cash & cash Equivalents

